Milestone 3 : SQL Refactoring Assistant

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Overview

Goal

Design, justify and verify model architecture

Built end-to-end pipeline using **T5**

Successfully tested on subset of data

Verified

→ **Preprocessing**

→ **Tokenization**

→ **Model generation**

→ **Output decoding**

Dataset Organisation

Structured directories for modular workflow

/data/raw/

Original dataset

/data/processed/

Cleaned SQL pairs

/data/splits/

Train / Validation / Test

Dataset Split (668 samples)

466	100	101
Train	Validation	Test

Preprocessing & Tokenization

Preprocessing

- Syntax validation using `sqlglot`
- Lowercasing & whitespace cleanup
- Removed duplicates & invalid queries

Tokenization

Prefix: "translate sqlite to hive:"

T5Tokenizer with:

- Max length: **128**
- Padding & truncation

Architecture Description

MODEL: T5 (T5-SMALL)

ENCODER-DECODER TRANSFORMER

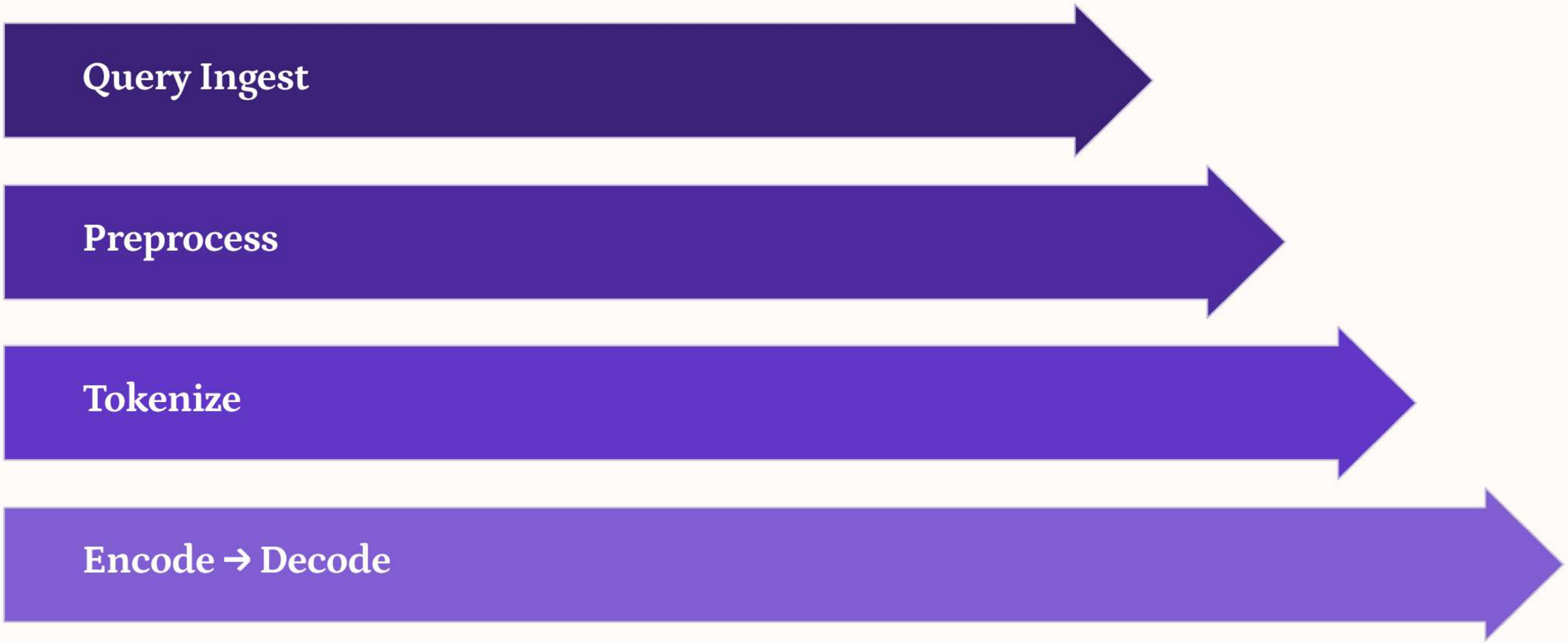
Encoder

- Processes input query bidirectionally
- Creates contextual representation

Decoder

- Generates HiveQL step-by-step
- Uses cross-attention with encoder output

Data Flow Diagram



The pipeline transforms a raw SQLite query through preprocessing, tokenization, and the T5 encoder-decoder architecture before producing the final HiveQL output.

Input Format & Tensors

Fixed Tensor Shape

Batch Size: **128**

Format: **PyTorch tensors**

Supports **GPU (CUDA)** or **CPU**

Inputs

`input_ids`

Token ID sequences for the input query

`attention_mask`

Masks padding tokens during processing

Suitability of Architecture

T5 treats tasks as **text-to-text** → perfect for SQL translation

Why T5?

Captures full input context

Bidirectional encoder understands the entire SQLite query before generating output.

Ensures structured output generation

Decoder produces well-formed HiveQL token by token.

Better control than decoder-only models

Encoder-decoder separation gives finer control over translation quality.

Efficient even with smaller variant (t5-small)

Achieves strong results without requiring large compute resources.

Pipeline Verification

Tested on 10 samples

Result

- No errors
- Stable and working pipeline

VALIDATED:

01

Preprocessing

02

Tokenization

03

Model inference

04

Output decoding

Outputs, Loss & Metrics

Outputs

Correct SQL structure

Partial HiveQL accuracy

(no fine-tuning yet)

Loss Function

Cross-Entropy Loss

Teacher forcing

Ignores padding

Evaluation

78%

Similarity Score

Future Work

- CodeBLEU
- Execution Accuracy
- Model fine-tuning